Efficient Randomized Experiments Using Foundation Models

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Problem Setting

- We collect an experimental sample $(A_i, X_i, Y_i)_{i=1}^n$ with known treatment probability $\pi = \mathbb{P}(A = 1)$ where A are treatment variables, X the covariates, Y outcomes
- We want to estimate average treatment effect

$$\theta = \mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0]$$

- **Standard practice**: Estimate θ via the *unbiased* AIPW estimator

$$\hat{\theta}(\hat{h}) = \sum_{i=1}^{n} \frac{A_i - \pi}{\pi(1-\pi)}(Y_i - \hat{h}(X_i, A_i)) + \hat{h}(X_i, 1) - \hat{h}(X_i, 0)$$

- But the outcome model $\hat{h}$ is typically learned from a small experimental sample using e.g. linear regression

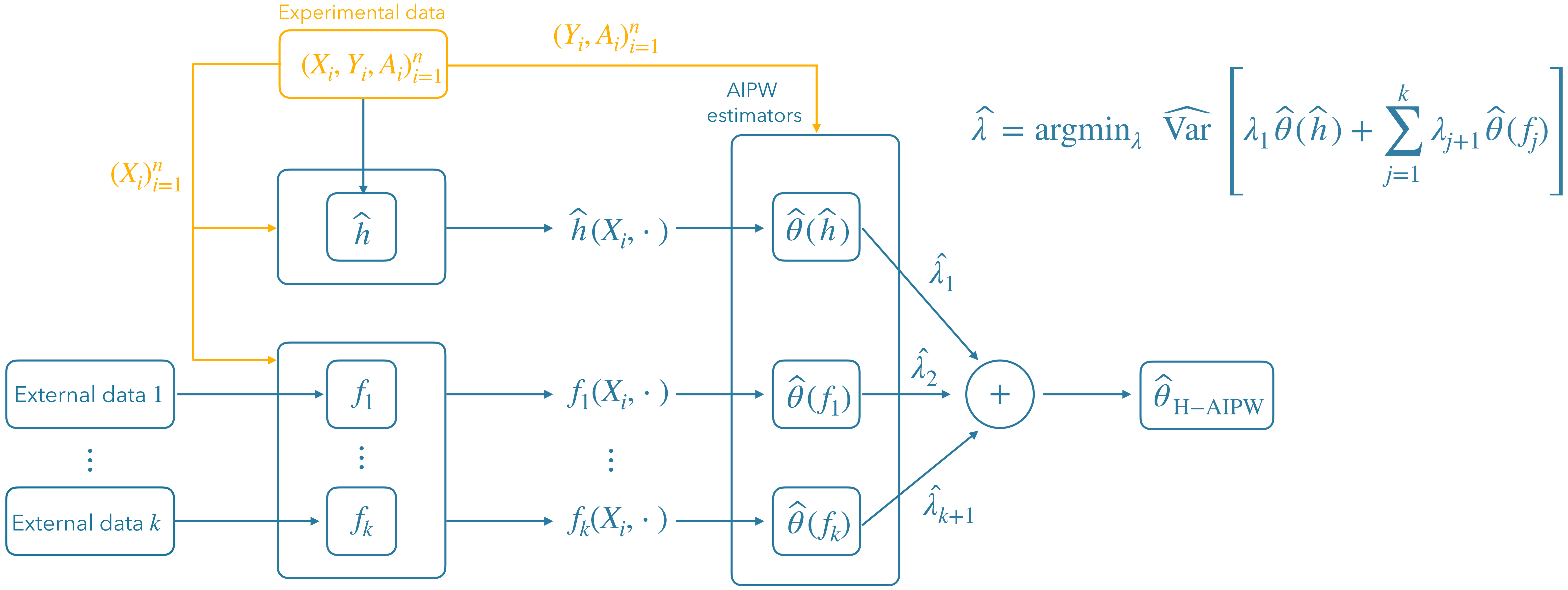
We propose the **H**ybrid **A**ugmented **I**nverse **P**robability **W**eighting **(H-AIPW)**, a novel estimator that can safely leverage multiple foundation models to improve efficiency.

☑ Protects against bias from the foundation models

☑ Always **at least as efficient** as the classic AIPW estimator

Method

- Use large external datasets to learn different, more expressive outcome models f_i
- Then compute separate AIPW estimators $\hat{\theta}(f_i)$ for each such model and combine them
- No matter how biased f_i themselves are, **all $\hat{\theta}(f_i)$ are unbiased** in expectation over the experimental data!



Experiments using LLMs on survey data

Example setting: Effect of protest framing on cancel culture opposition
Translating covariates x_i and treatment = 1 (blue) to an LLM prompt:

```
You are a 35-year-old female Democrat with liberal views and $75k
income.  A university banned an Antifa protest citing safety concerns
and that such protests contribute to cancel culture.
How much do you agree:  "Cancel culture is a big problem in today's
society"?  (1-5 scale)
```

Recorded integer value output $f(x_i, 1)$ for a trained language model f

> As someone with liberal views, I'm skeptical of the university's framing. While I believe in maintaining safety, linking Antifa protests to "cancel culture" seems politically motivated.
> I'd rate my agreement as 2 - Disagree.

Comparison to methods: Scaled variance of several estimators, smaller values indicate better precision.
The blue row is the experimental data only baseline that all the estimators should improve upon.
Importantly, our estimator is guaranteed to achieve precision at least as good as this baseline.

	Melin et al. (2022)		Silverman et al. (2022)		Kennedy et al. (2020)		Fahey et al. (2023)	
Estimator	$n = 100$	$n = 200$	$n = 100$	$n = 200$	$n = 100$	$n = 200$	$n = 100$	$n = 200$
H-AIPW	**10.39**	**10.28**	**2.10**	**2.14**	**17.09**	**17.47**	**4.87**	4.94
PPI-style	11.00	11.06	2.25	2.26	17.87	17.97	4.88	**4.91**
PROCOVA	11.81	10.62	2.24	2.22	18.38	18.11	5.18	5.09
AIPW (boosting)	12.82	12.44	2.82	2.83	23.09	23.12	6.31	6.37
AIPW (standard)	11.72	10.57	2.22	2.20	18.09	17.95	5.09	5.04
DM	11.10	11.10	2.30	2.30	18.07	18.08	5.61	5.62